OCD-fold family (Sus): full maximum-likelihood tree
125 sequences; 125 sequences with 434 amino-acid sites; model Q.PFAM+R5; node labels SH-aLRT (%) / UFBoot (%), shown when UFBoot ≥ 50; rooted as in Fig. 3
Tylopilus sp. SUS_JBBPWX010000117_L1 [pseudogene-like]
Tylopilus sp. SUS_JBBPWX010003679_L1 [pseudogene-like]
Tylopilus sp. SUS_JBBPWX010001246_L1 [pseudogene-like]
Tylopilus sp. SUS_JBBPWX010003574_L1 [pseudogene-like]
Neotropicomus parvogracilis SUS_JBBPJX010000484_L1 [pseudogene-like]
Neotropicomus parvogracilis SUS_JBBPJX010000568_L1 [pseudogene-like]
Nevesoporus exiguus SUS_JBBPKI010000869_L1 [intact]
Nevesoporus exiguus SUS_JBBPKI010000725_L1 [pseudogene-like]
Nevesoporus exiguus SUS_JBBPKI010001675_L1 [pseudogene-like]
Nevesoporus exiguus SUS_JBBPKI010000406_L1 [pseudogene-like]
Nevesoporus exiguus SUS_JBBPKI010000567_L1 [intact]
Boletus variipes SUS_JBBPJM010005915_L1 [intact]
Austroboletus festivus SUS_JBBPOG010000890_L1 [intact]
Veloporphyrellus conicus SUS_JBBPLJ010000677_L1 [intact]
Veloporphyrellus conicus SUS_JBBPLJ010000841_L1 [intact]
Boletellus deceptivus SUS_JBBPMG010004664_L1 [intact]
Boletellus deceptivus SUS_JBBPMG010004664_S2L1 [intact]
Boletellus deceptivus SUS_JBBPMG010001931_L1 [intact]
Boletellus ananiceps SUS_JBBPLV010001121_L1 [intact]
Xerocomus nothofagi SUS_JBBPLY010003693_L1 [intact]
Veloboletus limbatus SUS_JBBPJJ010005399_L1 [intact]
Boletus reticuloceps SUS_JAGFBS010000039_L1 [pseudogene-like]
Fistulinella staudtii SUS_JBBPTS010002895_L1 [pseudogene-like]
Tylopilus sp. SUS_JBBPWX010001359_L1 [pseudogene-like]
Boletus rex-veris SUS_JBBPJQ010000015_L1 [pseudogene-like]
Boletus rex-veris SUS_JBBPJQ010001889_L1 [pseudogene-like]
Melanogaster broomeanus SUS_WIOP01000636_L1 [pseudogene-like]
Rubroboletus satanas SUS_JBBPNB010002703_L1 [pseudogene-like]
Gastroboletus turbinatus SUS_JBBPND010000159_L1 [pseudogene-like]
Chalciporus aff. SUS_JBBPJZ010002221_L1 [pseudogene-like]
Chalciporus aff. SUS_JBBPJZ010007258_L1 [intact]
Boletellus aff. SUS_JBBPTT010006837_L1 [intact]
Phylloporus sp. SUS_JBBPVF010000849_L1 [intact]
Phylloporus sp. SUS_JBBPWB010005531_L1 [intact]
Boletellus emodensis SUS_JBBPLU010000323_L1 [intact]
Boletellus deceptivus SUS_JBBPMG010000116_L1 [intact]
Boletellus ananiceps SUS_JBBPLV010001943_L1 [intact]
Boletellus ananiceps SUS_JBBPLV010001943_S2L1 [intact]
Phylloporus aff. SUS_JBBPUD010000148_L1 [intact]
Porphyrellus sp. SUS_JBBPVZ010000240_L1 [intact]
Porphyrellus sp. SUS_JBBPVZ010030891_L1 [intact]
Tylopilus aff. SUS_JBBPTQ010005164_L1 [intact]
Boletaceae sp. SUS_JBBPWO010009350_L1 [intact]
Boletaceae sp. SUS_JBBPWO010073651_L1 [intact]
Porphyrellus sp. SUS_JBBPXK010000119_L1 [intact]
Boletaceae sp. SUS_JBBPWT010000047_L1 [intact]
Tylopilus sp. SUS_JBBPXA010000488_L1 [intact]
Porphyrellus sp. SUS_JBBPWU010000114_L1 [pseudogene-like]
Porphyrellus sp. SUS_JBBPXF010000575_L1 [intact]
Boletellus sp. SUS_JBBPXB010001911_L1 [intact]
Boletellus sp. SUS_JBBPXB010007365_L1 [intact]
Mortierella alpina SUS_JANATF010000012_L1 [pseudogene-like]
plasmid: WP_107028680.1 hypothetical protein Ensifer sp NM-2
plasmid: WP_269710210.1 hypothetical protein Rhizobium rhizogenes
WP_174080856.1 Rhizobiaceae
WP_174075806.1 Martinezella rhizogenes
plant cT-DNA: Ipomoea batatas cultivar Taizhong6 chromosome 7 chr7 who
plant cT-DNA: Diospyros lotus cultivar W01 chromosome 8 whole genome s
plant cT-DNA: Juglans mandshurica isolate NFU 2021 chromosome 7 whole
plant cT-DNA: Juglans cathayensis isolate DJUG11 03 scaffold974 whole
plant cT-DNA: Juglans sigillata isolate DJUG951 04 scaffold4680 whole
plant cT-DNA: Cyclocarya paliurus isolate 4PA2023 chromosome 3D whole
plasmid: WP_034521016.1 MULTISPECIES hypothetical protein Rhizobium Ag
plasmid: WP_174158420.1 hypothetical protein Rhizobium rhizogenes
plant cT-DNA: Quillaja saponaria isolate S10 chromosome 2 whole genome
plant cT-DNA: Parasponia andersonii isolate WU1-14 PanWU01x14 asm01 sc
plant cT-DNA: PON40587 1 Parasponia andersonii extra
plant cT-DNA: PON40593 1 Parasponia andersonii extra
plant cT-DNA: Nicotiana tomentosiformis chromosome 7 whole genome shot
plant cT-DNA: Nicotiana tabacum whole genome shotgun sequence AWOK0161
plant cT-DNA: Diospyros kaki Taishu DNA chromosome 4 draft genome sequ
plant cT-DNA: PON40591 1 Parasponia andersonii extra
plant cT-DNA: Trema orientale isolate RG33-2 TorRG33x02 asm01 scf00104
plant cT-DNA: Parasponia andersonii isolate WU1-14 PanWU01x14 asm01 sc
plant cT-DNA: Euphorbia esula cultivar 1984-ND001 c294543 g1 i1 whole
plasmid: WP_173994934.1 MULTISPECIES hypothetical protein Rhizobium Ag
WP_210057129.1 Neorhizobium petrolearium
WP_233954827.1 Neorhizobium galegae
WP_085857703.1 unclassified Rhizobium
WP_084365946.1 Rhizobium sp. RU36D
WP_069042703.1 unclassified Agrobacterium
WP_026615224.1 Ensifer aridi
WP_026619421.1 Ensifer sp. WSM1721
WP_037418621.1 Sinorhizobium sp. CCBAU 05631
WP_097540713.1 Sinorhizobium
outgroup: KAH0437978.1 Aureobasidium melanogenum
outgroup: XP 056037437.1 Schizosaccharomyces osmophilus
PZN56776.1 Pseudomonadota bacterium
WIE93331.1 Mesorhizobium sp. WSM4875
WP_126097880.1 Mesorhizobium sp. M1B.F.Ca.ET.045.04.1.1
WP_206548719.1 Alphaproteobacteria
WP_228017358.1 Phyllobacterium pellucidum
WP_262294504.1 Phyllobacterium zundukense
outgroup: WP_151986804.1 Rhizobium sp. EC-SD404
QOT73213.1 Sphingobium fuliginis
outgroup: WP_062264854.1 Methanoculleus bourgensis
outgroup: WP_089777055.1 Haloferax massiliensis
BFD90164.1 Kitasatospora sp. Xyl93
WP_313720880.1 Micromonospora halotolerans
APF35841.1 Microbacterium paludicola
HYH28366.1 Actinomycetota bacterium
TMC60602.1 Chloroflexota bacterium
outgroup: KAF8552678.1 Imleria badia
outgroup: KFH71693.1 Podila verticillata NRRL 6337
outgroup: WP_188602682.1 Vulcanisaeta souniana
WP_127491873.1 Paenibacillus glycanilyticus
WP_329068487.1 Streptomyces sp. NBC 01429
outgroup: XP 007840637.1 Pestalotiopsis fici W106-1
native Pseudopestalotiopsis theae SUS_JACWGD010000003_L1 [intact]
native Pseudopestalotiopsis chinensis SUS_JAPTGH010000011_L1 [intact]
WP_114836055.1 Agrobacterium tumefaciens
UXT93026.1 Agrobacterium pusense
AUC13453.1 Rhizobium sp. Y9
QDG94046.1 Rhizobium sp. NIBRBAC000502774
WP_262165449.1 Devosia neptuniae
MFA7416147.1 Rhizobium sp.
MBA3043409.1 Rhizobiaceae bacterium
WP_085545780.1 Burkholderia pseudomallei
WP_250131643.1 Salmonella bongori
WP_188326420.1 Stenotrophomonas sp. 169
Boletus speciosus SUS_QFEG01000233_L1 [intact]
Retiboletus ornatipes SUS_QFEH01000351_L1 [intact]
Echinosphaeria canescens SUS_JAOXNW010100945_L1 [intact]
outgroup: plasmid: WP_111794418.1 MULTISPECIES hypothetical protein Ag
Austroboletus olivaceoglutinosus SUS_JBBPSN010000183_L1 [intact]
0.5 substitutions/site
T-DNA sus clade (fungal sus-like + Ti/Ri plasmid and plant cT-DNA references) (76)
Rhizobiales chromosomal OCD (19)
other bacterial OCD-fold proteins (14)
other fungal OCD-like (4)
native fungal homologs (2)
outgroup (10)